

Organism and Population

1. Select the correct statement w.r.t. ecological niche

- (1) Concept of ecological niche was given by Darwin.
- (2) Ecological niche does not represent the range of conditions an organism can tolerate
- (3) It describes the functional role of an organism
- (4) Two species can occupy exactly same niche

2. Read the following Assertion (A) and Reason (R) statements and choose the correct option.

Assertion (A) : In interference competition, even if resources are abundant, the feeding efficiency of both the species increases due to inhibitory presence of the other species.

Reason (R) : Resources must be limiting in any habitat for competition to occur.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

3. Which one of the following is not a parasitic adaptation?

- (1) Development of adhesive organs
- (2) Loss of digestive organs
- (3) Loss of reproductive capacity
- (4) Loss of unnecessary sense organs

4. What percentage of PAR is captured by plants?

- (1) 100%
- (2) 50%
- (3) 2-10%
- (4) 1-5%

5. Match the following columns

Column I	Column II
a. Predation	(i) Clown fish-Sea anemone
b. Competition	(ii) Sparrow-seed
c. Commensalism	(iii) Sea anemone-Hermit crab
d. Protocooperation	(iv) Balanus-Chathamalus
(1) a(ii), b(iv), c(i), d(iii)	
(2) a(ii), b(i), c(iv), d(iii)	
(3) a(iii), b(ii), c(iv), d(i)	
(4) a(iii), b(iv), c(i), d(ii)	

6. Ecology is basically concerned with four level of biological organisation. These are :

- (1) Biosphere, population, habitat, community
- (2) Population, habitat, organism, community
- (3) Organisms, population, communities, biome
- (4) Habitat, population, biome, organism

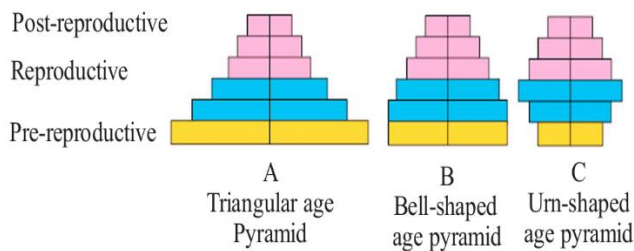
7. Identify the correct set of statements :

- (a) Ramedeo Misra is known as father of ecology in India.
- (b) Mean annual precipitation in grassland is more than 200 cm.
- (c) Annual variation in intensity and duration of light on earth is due to its rotation around the sun and the tilt of its axis.
- (d) Tropical forest shows the temperature variation in a range between 0°C to 30°C.
- (e) Regional and local variation within biome lead to the formation of a wide variety of habitats.

Choose the correct answer from options

- (1) a, b, c and e
- (2) a, b and c
- (3) a, c and e
- (4) a, c, d and e

8. Age pyramid A, B and C indicates :



- (1) A - Expanding population, B - Stable population
C - Declining population
- (2) A - Expanding population, B - Declining
population 1 C - Stable population
- (3) A - Stable population, B - Declining population C
- Expanding population
- (4) A - Declining population, B - Stable population C
- Expanding population

9. Given below are two statements:

Statement-I : Immigration is the number of individuals of same species who left the habitat and gone elsewhere during the time period under consideration.

Statement-II : Natality and emigration contribute to an increase in population density.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and II both are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

10. Type of population interaction found between visiting flamingoes and resident fishes in South American lakes is

- (1) Predation
- (2) Competition
- (3) Parasitism
- (4) Commensalism

11. The mean annual precipitation will be highest in

- (1) Coniferous forest
- (2) Temperate forest
- (3) Grassland
- (4) Tropical rain forest

12. Given below are two statements;

Assertion (A) : The interaction where one species is benefited and the other is neither benefited nor harmed, is called commensalism.

Reason (R) : An orchid growing as an epiphyte on a mango branch is an example of commensalism.

Choose the most appropriate answer

- (1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is wrong.
- (3) (A) is incorrect but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

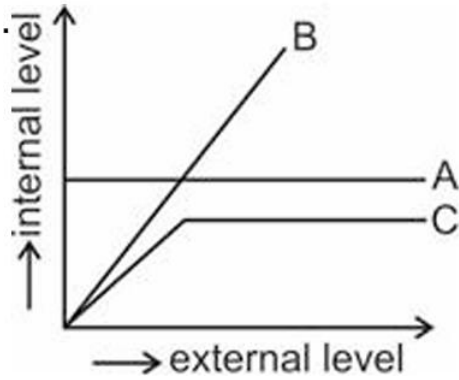
13. Which of the following statement is incorrect :

- (1) In the aquatic environment the sediment characteristics often determine the type of benthic animals that can thrive there
- (2) In animal the organism if unable to migrate, might avoid the stress by escaping in time
- (3) Many species of small plants growing in forests are adapted to photosynthesis optimally under very high light conditions
- (4) The levels of thermal tolerance of different species determine their geographical distribution to a large extent

14. If in a pond there are 48 fishes last year and through reproduction 10 new fishes are added taking the current population to 58. What will be the birth rate pre fish per year :

- (1) 0.172
- (2) 0.827
- (3) 0.208
- (4) 4.8

15.



Given above is diagrammatic representation of organismic response. Identify the correct statements for A, B and C.

- (a) 'A' is applicable for birds and mammals
- (b) 'B' is true for organisms whose body temperature changes with atmosphere
- (c) 'C' is for those organisms which show osmoregulation but no thermoregulation.

- (1) Only a and b
- (2) Only b and c
- (3) Only a and c
- (4) All a, b and c

16. Mean annual temperature in a grassland ecosystem

- (1) Is more than that of arctic and alpine tundra
- (2) Does not exceed 10°C
- (3) Does not go below 15°C
- (4) Goes below 0 degree Celsius always

17. A population 'X' was reported with the following data.

- a. Number of pre-reproductive individuals = 9,875
- b. Number of reproductive individuals = 18,960
- c. Number of post-reproductive individuals = 11,500

Which of the following conclusions can be withdrawn from the above data?

- (1) Such population shows bell-shaped pyramid
- (2) Above population will decline with time
- (3) This is a stable population
- (4) This population will keep on growing always

18. Identify the incorrect set of statements :

- (a) Predators help in maintaining species diversity in a community.
- (b) Predators in nature are prudent.
- (c) The weed calotropis produces highly poisonous cardiac glycosides.
- (d) The monarch butterfly is highly distasteful to its predator because of a special chemical present in its body.
- (e) A predator species is sustainable when it is highly efficient.

Choose the correct answer from options

- (1) a, b and e
- (2) a, b, c and d
- (3) d and e
- (4) only e

19. Match the Column-I with Column-II

(a)	Saprophyte	(i)	Symbiotic association of fungi with plant roots
(b)	Parasite	(ii)	Decomposition of dead organic materials
(c)	Lichens	(iii)	Living on living plants or animals
(d)	Mycorrhiza	(iv)	Symbiotic association of algae and fungi

- (1) a(ii), b(iii), c(iv), d(i)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(ii), b(i), c(iii), d(iv)

20. Very small animals are rarely found in polar regions because of:

- (1) Presence of small heart
- (2) They get buried due to snowfall
- (3) RBCs becoming unable to transport oxygen
- (4) They have large surface area relative to their volume

21. Find the odd one w.r.t. homeostasis.

- (1) Majority of animals and nearly all plants cannot maintain homeostasis
- (2) Aquatic animals are conformers
- (3) Very few lower vertebrate and invertebrate species are able to maintain homeostasis by physiological and behavioural means.
- (4) All birds and mammals are conformers

22. _____ is an edaphic factor.

Select the correct answer to fill in the blank.

- (1) Water
- (2) Light
- (3) Temperature
- (4) Soil

23. In a newly colonized habitat, which of the following factors contributes more significantly to population growth?

- (1) Births
- (2) Deaths
- (3) Immigration
- (4) Emigration

24. A population has certain attributes. This attributes are :

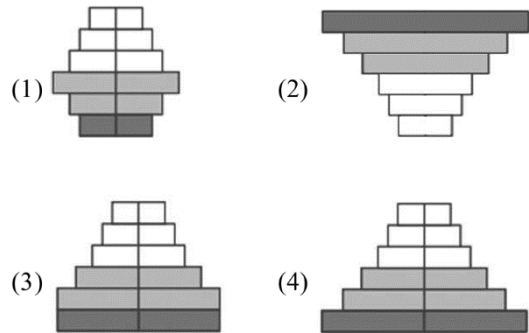
- a. Death b. Birth
- c. Death rate d. Birth rate
- e. Sex ratio

- (1) a, b, c, d, e
- (2) c, d, e
- (3) Only e
- (4) Only c, d

25. A pyramid with pre-reproductive individuals being more than reproductive individuals the population is said to bea..... withb.....age pyramid :

- (1) a- Declining b- Urn shaped
- (2) a- expanding b-Triangular
- (3) a- stable b- bell shaped
- (4) a- expanding b- Bell shaped

26. Which of the given age pyramid reflects an increasing human population:



27. Which of the following statements are correct w.r.t population :

- (A) Some species of insects and frogs are cryptically coloured to avoid being detected easily by the predator
- (B) Lichens represent an intimate mutualistic relationship between a fungus and photosynthesising algae
- (C) CAM plants morphologically adapt to open stomata during daytime

Select the correct answer

- (1) Only A, C is correct
- (2) Only B, C is correct
- (3) Only C is incorrect
- (4) All are correct

28. Asymptote in a logistic growth curve is obtained when :

- (1) $K > N$
- (2) $K < N$
- (3) The value of 'r' approaches zero
- (4) $K = N$

29. Which of the following is odd w.r.t. adaptations shown by parasites?

- (1) Presence of adhesive organs or suckers to cling onto the host.
- (2) High rate of reproduction
- (3) Presence of complex digestive system for digesting the food obtained from the host
- (4) Loss of unnecessary sense organs.

30. Select In the equation of logistic growth

$$\left[\frac{dN}{dt} = r_N \left(\frac{K - N}{K} \right) \right], \left(\frac{K - N}{K} \right)$$

- (1) influence of environmental resistance
- (2) intrinsic rate of natural increases
- (3) exponential form of growth
- (4) influence of individuals over biotic potential

31. Find out correct statement among the following :

- (a) The productivity and distribution of plants is also heavily dependent on water.
- (b) Some organisms are tolerant of a wide range of salinities called euryhaline while others are restricted to a narrow range called stenohaline.
- (c) Mammals of colder climates generally have shorter ears and limbs to minimise heat loss, this is called the Allen's Rule.
- (d) If the age distribution of different members in population is plotted, the resulting structures is called an average community pyramid

Select the correct answer

- (1) only (a) & (b)
- (2) only (c) & (d)
- (3) only (a), (b) & (c)
- (4) (a), (b), (c) & (d)

32. What is correct about amensalism :

- (1) One species is harmed and other is unaffected
- (2) One species is harmed and other is benefitted
- (3) One species in benefitted and other is neither benefitted nor harmed
- (4) One species is harmed and other is partially benefitted.

33. If a new habitat is just being colonised which process may contribute more significantly to population growth :

- (1) Birth rates
- (2) Immigration
- (3) Emigration
- (4) Mortality

34. The logistic population growth is expressed by the equation :

- (1) $dt / dN = Nr \left(\frac{K - N}{K} \right)$
- (2) $dN / dt = rN \left(\frac{K - N}{K} \right)$
- (3) $dN/dt = rN$
- (4) $dN / dt = rN \left(\frac{N - K}{K} \right)$

35. Select the odd one w.r.t. altitude sickness

- (1) It can be experienced at high altitude, where body does not get enough oxygen due to low atmospheric pressure.
- (2) Causes nausea, fatigue and heart
- (3) Body acclimatise this sickness by increasing RBC production, increasing binding affinity of haemoglobin to oxygen and increasing breathing rate
- (4) Organisms acclimatise themselves by physiological means.

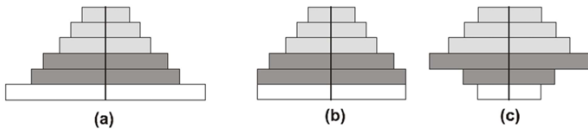
36. Read the following statements regarding adaptations.

- (a) Organisms living at great depths in the ocean show a fascinating array of biochemical adaptations.
- (b) Xerophytes have sunken stomata and thin cuticle on their leaf surface to minimize the loss of water through transpiration
- (c) Mammals from colder climate have shorter ears and long limbs to minimise heat loss
- (d) Kangaroo rat is capable of meeting all its food requirement through internal protein oxidation only.

Select the correct one(s).

- (1) (a), (c) and (d) only
- (2) (b), (c) and (d) only
- (3) (a) and (c) only
- (4) (a) only

37.



Above diagram represents age pyramids of human population. In which a, b & c represent triangular, Bell shaped and Urn shaped age pyramids which of the following statement is true

- (1) The number of pre-reproductive individuals is more than reproductive individuals in Bell shaped age pyramids
- (2) The number of post reproductive individuals is equal to number of reproductive individuals in Triangular age pyramid
- (3) The number of reproductive individuals is more than post reproductive individuals and pre reproductive individuals in urn shaped age pyramid
- (4) The number of pre-reproductive individuals & reproductive individuals are equal in triangular shaped curve

38. Read the given statements

Assertion :- Competition always occurs between closely related species compete for same resources that are limiting.

Reason :- Totally unrelated species could not compete for same resources.

Select the correct answer

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

39. Which one is the incorrect match:

- (1) Mutualism - ++
- (2) Competition - --
- (3) Predation - --
- (4) Parasitism - +-

40. In nature in Interspecific competition, they point out that species facing competition might evolve mechanism that promote co-existence rather than exclusion. One such mechanism is :

- (1) Parasitism
- (2) Resource partitioning
- (3) Commensalism
- (4) Amensalism

41. A stable human population is reflected by

- (1) Triangular shaped age pyramid
- (2) Bell shaped age pyramid
- (3) Urn shaped age pyramid
- (4) Inverted age pyramid

42. Stage of suspended development, which is found in many zooplankton species in lakes and ponds under unfavourable condition is known as

- (1) Migration
- (2) Diapause
- (3) Aestivation
- (4) Hibernation

43. Read following statements w.r.t. growth curves?

- (A) A population of elephants in a forest with limiting resources will show a S-shaped growth curve.
- (B) The influence of environmental resistance over biotic potential is denoted by $(K - N/K)$
- (C) When $K = N$, asymptote in a logistic growth curve is obtained.
- (D) Value of r is an important parameter to assess the impact of population growth on the environment.
- (E) Decline in exponential growth of a population is density triggered.

Select the correct answer

- (1) (B), (D) and (E) are false
- (2) (A), (B), (C) and (E) are true
- (3) (B) and (D) are false
- (4) Only (D) is true

44. Match the following :

(i)	Regulation	(a)	Body temperature changes as per environment
(ii)	Conforming	(b)	Maintain homeostasis
(iii)	Migration	(c)	Temporary movement
(iv)	Suspension	(d)	Escape unfavourable condition

- (1) i-b, ii-c, iii-d, iv-a
- (2) i-a, ii-b, iii-c, iv-d
- (3) i-b, ii-a, iii-c, iv-d
- (4) i-a, ii-d, iii-b, iv-c.

45. Mark the incorrect statement w.r.t. predation

- (1) Herbivores have a greater advantage since plants cannot run away to avoid predation
- (2) If a predator is prudent, the prey population will become extinct
- (3) Predators help in maintaining species diversity
- (4) Predation keep prey population under control

46. Read the below given statements and select the correct option.

Statement-A : Most of the mammals and birds are eurythermal.

Statement-B : Polar bears, Lizards, Abies are stenothermal.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

47. Calotropis growing in abandoned field produces highly poisonous chemical, which protects it from grazing herbivores. The chemical is called

- (1) Nicotine
- (2) Caffeine
- (3) Cardiac glycosides
- (4) Strychnine

48. Which of the following is incorrect w.r.t. methods for determining population size?

- (a) Estimation of grasses in a forest using total counts.
- (b) Percentage cover is an absolute method of determining population size.
- (c) Bacterial growth in a liquid culture can be estimated using relative densities.
- (d) Population of tiger can be estimated based on the pug marks and fecal pellets.

- (1) (a) and (c) only
- (2) (b) and (d) only
- (3) (c) and (d) only
- (4) (a) and (b) only

49. Which ones are true about interdependence.

- (a) One is benefitted and other unaffected in mutualism.
- (b) Both partners are benefitted in commensalism.
- (c) One kills and feeds on another in predation
- (d) Both partners are benefitted in mutualism

Select the correct option.

- (1) a and b only
- (2) a and c only
- (3) b and c only
- (4) c and d only

50. What will be the estimated population size of the tigers in a forest, where there were initially 30 tigers and in a span of 1 year, 14 new tigers were born, of which 8 died due to malnutrition and 4 nursing mothers also died. At the same time, 10 new tigers immigrated from the adjoining forests. What will be the final tiger count in the forest?

- (1) 22
- (2) 42
- (3) 32
- (4) 30

Answer Key

1. (3)
2. (4)
3. (3)
4. (3)
5. (1)
6. (3)
7. (3)
8. (1)
9. (1)
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43. (2)
44. (3)
45. (2)



- 46. (3)
- 47. (3)
- 48. (4)
- 49. (4)
- 50. (2)

